

AMT Horizon

Functional Specification Document

DOCUMENT VERSION 1.2

14/01/2026

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Revision History

Date	Version	Description of Changes
04/01/2026	1.0	- Document creation and drafting of sections Overview, A to E. - Added Functional Requirements.
08/01/2026	1.1	- Translated code and technical sections to English. - Adapted for NextJS, Vitest, Neon, Drizzle, better-auth.
14/01/2026	1.2	- Separated Functional and Technical Specifications into distinct documents. - Added Page Functionality section. - Updated Functional Requirements with detailed descriptions.

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I. General Overview

Our client requested the development of a web/mobile application in NextJS with better-auth to manage taxi rides, their distribution, and billing. The goal is to provide an intuitive solution for admins, drivers, and customers, with advanced features such as data import via photo (OCR) and automatic ride suggestions.

A. Product Description

This application allows you to:

- **Cataloging** taxi rides.
- **Distributing** rides among drivers via a dynamic schedule.
- **Generating** invoices from rides (individual or grouped).
- **Suggesting** rides to drivers and allowing them to request assignment.
- **Importing** ride data via form, text, schema, or photo (OCR).

B. Product Functional Capabilities

User Management

- **Registration/Login:** Distinct roles (Admin, Driver, Customer) with secure authentication (better-Auth).
- **Password Recovery:** Reset link valid for 24 hours.

Ride Management

- **Manual Addition:** Form to enter details (date, time, location, price).
- **Import via Text/Photo:** Automatic data extraction via OCR (Tesseract.js/Google Vision).
- **Ride Suggestions:** Matching algorithm based on availability and location.
- **Assignment Request:** Drivers can request to be assigned to a ride.

Admin View (Excel-like)

- **Dynamic Table:** Filters, sorting, data export.
- **Export:** Generate CSV/PDF files.

Billing

- **Invoice Creation:** Select rides, automatic numbering.
- **Invoice Sending:** By email, with status tracking.

Notifications

- **Real-time Alerts:** New rides, assignments, invoices.

Additional Features

- **Geolocation:** Integration with Mapbox/Google Maps.
- **Online Payment:** Stripe/PayPal.
- **Internal Chat:** Communication between drivers and customers.

C. Project Organization

Project Representatives

Project Owner	Represented by...
Client	Alexis SANTOS (Program Manager)

Stakeholders

Stakeholder	Interest
Admins	Full management of rides and invoices.
Drivers	View suggested rides, request assignment.
Customers	Add rides, track trips.

II. Requirements

A. Functional Requirements

User Management

ID	Feature	Actors	Inputs	Outputs	Business Rules
FR-001	Registration/Login	All	Email, password, role	Account created, active session	Roles: Admin, Driver, Customer
FR-002	Password Recovery	All	Email	Reset link	Link valid for 24h
FR-003	Driver Account Validation	Admin	ID card, driver's license	Account validated or rejected	Manual verification within 24h.
FR-004	Profile Modification	All	New email, phone number	Profile updated, notification	Unique email, valid format.
FR-005	Account Deactivation	Admin	User, Reason (optional)	Account deactivated	Account disabled, notification, Data archiving (GDPR).

Gestion des Courses

ID	Fonctionnalité	Acteurs	Entrées	Sorties	Règles Métier
FR-010	Manual Ride Addition	All	Form (date, time, location, price)	Ride recorded, notification	Required fields validated
FR-011	Ride Import via Photo	Admin, Drivers	Photo (ticket, invoice)	Extracted data, ride created	OCR + manual validation
FR-012	Ride Suggestions	System	Availability, location	List of suggested rides	Matching algorithm
FR-013	Manual Validation of OCR Imports	Admin	Extracted data, manual correction	Ride validated or rejected	Notification required for rejection (reason mandatory).
FR-014	Ride History	Drivers, Customers	Filters (date, status)	List of past rides	Customers only see their own rides.
FR-015	Ride Cancellation	Drivers, Customers	Reason, ride ID	Ride cancelled, notification	Max delay: 2 hours before departure.

ID	Fonctionnalité	Acteurs	Entrées	Sorties	Règles Métier
FR-016	Driver Rating	Customers	Rating (1-5), comment	Rating recorded, average updated	Only after a completed ride.
FR-017	Automatic Ride Suggestions	System	Location, availability	List of 3 priority rides	Algorithm based on proximity and history.
FR-018	Assignment Request with Comment	Drivers	Ride ID, message	Request sent, admin notification	Max 5 simultaneous requests.

Admin View (Excel-like)

ID	Feature	Actors	Inputs	Outputs	Business Rules
FR-020	Dynamic Table	Admin	Filters (date, status)	Sortable/exportable list	Access restricted to admins
FR-021	Custom Export	Admin	Selected columns	CSV/PDF file with filtered data	Limit: 10,000 rows per export.
FR-022	Ride Statistics	Admin	Period, data type	Charts (number, revenue, etc.)	Automatic daily refresh.

Billing

ID	Fonctionnalité	Acteurs	Entrées	Sorties	Règles Métier
FR-030	Invoice Creation	Admin	Ride selection	Generated PDF invoice	Automatic numbering
FR-031	Grouped Billing	Admin	Ride selection	Single PDF invoice	Grouping by customer or period.
FR-032	Automatic Reminder	System	Overdue due date	Reminder email	Max 3 reminders (D+7, D+14, D+30).
FR-033	Exceptional Discount	Admin	Invoice ID, percentage	Updated invoice	Justification required.
FR-034	Send Invoice via App	Admin	Invoice ID, recipient	Invoice sent (PDF attached), status updated	Immediate sending, history preserved. Notifies customer via email and in-app.
FR-035	Manual Reminder Management	Admin	Invoice ID, custom message	Reminder sent (email/app), status updated	Only after explicit admin validation. Max 3 reminders (configurable).

Notifications

ID	Feature	Actors	Inputs	Outputs	Business Rules
FR-040	Real-time Notifications	All	Event (ride, invoice)	Email/in-app alert	Channel customization
FR-041	Channel Customization	All	Preferences (email, app)	Channels updated	Email by default, can be disabled.
FR-042	Urgent Notifications	System	Critical event	Push + email alert	Ex: ride cancellation by customer.

Additional Features

ID	Feature	Actors	Inputs	Outputs	Business Rules
FR-050	Route Geolocation	Drivers, Customers	GPS coordinates	Interactive map	Mapbox/Google Maps integration
FR-051	Real-time Chat	Drivers, Customers	Text message	Saved conversation	History kept for 30 days.
FR-052	Live GPS Tracking	Customers	Ride ID	Driver position in real-time	Can be disabled by driver.
FR-053	Partial Payment	Customers	Partial amount	Invoice updated	Minimum 30% of total amount.
FR-054	Calendar Integration	Drivers	Google Calendar/Outlook link	Synchronized events	Bidirectional synchronization.

Security

ID	Feature	Actors	Inputs	Outputs	Business Rules
FR-060	Activity Log	Admin	Filters (user, action)	List of actions	6-month retention (GDPR).
FR-061	Two-Factor Authentication	All	SMS/email code	Access granted or denied	Mandatory for admins.

B. Non-Functional Requirements

- **OCR:** Minimum 90% accuracy for data extraction.
- **Performance:** Response time < 2s for critical actions.
- **Security:** Encryption of sensitive data (GDPR).

C. Personas

1. Business Manager (Schedule Management)

Name: Marc Lefèvre

Age: 55

Role: Manager of a small taxi fleet in Île-de-France

Main Goals:

- Organize drivers' and vehicles' schedules **without errors**.
- Find available drivers to cover requests, especially during peak hours.
- Reduce time spent manually managing schedules.

Needs:

- A **simple and visual** tool to:
 - See drivers' and vehicles' availability at a glance.
 - Assign rides quickly, even at the last minute.
 - Receive alerts in case of cancellations or delays.
- A feature to **motivate drivers** (e.g., bonuses for accepting urgent rides).

Frustrations:

- Spending **3-4 hours a day** managing schedules on paper and calling drivers one by one.
 - Drivers who don't respond or forget to confirm their availability.
-

2. General and Administrative Secretary

Name: Nathalie Dubois

Age: 47

Role: Administrative secretary in an SME/PME

Main Goals:

- Manage supplier and customer invoices **without delay**.
- Create and send invoices quickly and without errors.
- Track payments and remind late-paying customers.

Needs:

- An **automated system** to:
 - Generate and send invoices in one click (integration with ride data).
 - Track payments and send automatic reminders.
 - Centralize documents (invoices, contracts, purchase orders) to avoid losing them.
- A **clear and intuitive** interface to avoid data entry errors.

Frustrations:

- Invoices piling up and late payments affecting cash flow.
 - Time wasted searching for misplaced documents or recopying data.
 - Drivers forgetting to provide ride receipts.
-

3. Driver (Unmotivated but Opportunistic)

Name: Djamel Kebabti

Age: 39

Role: Independent taxi driver

Main Goals:

- **Maximize income** with minimal effort.
- Choose the **most profitable rides** (long distances, high fares).
- Avoid problematic customers or complicated routes.

Needs:

- An app that allows him to:
 - See available rides **with price and distance** before accepting.
 - Discreetly decline uninteresting rides.
 - Receive notifications for well-paid rides.
- A **customer rating system** (to avoid bad payers or unnecessarily long trips).

Frustrations:

- Having to call dispatch to get ride details.
 - Poorly paid rides or customers who change destinations en route.
 - Lack of transparency on fares and bonuses.
-

4. Classic Customer (Taxi Île-de-France)

Name: Jean-Michel Durand

Age: 62

Role: Retired, regular taxi user

Main Goals:

- Find a taxi **quickly and without complications**.
- Have a **punctual and professional** driver.
- Pay by cash or card without issues.

Needs:

- A **simple and reliable** solution to:
 - Order a taxi by phone or via a basic app.
 - Know the price in advance (no surprises).
 - Have a driver who knows the area well.

Frustrations:

- Long wait times during peak hours.
 - Drivers who don't know shortcuts or traffic jams.
 - Apps that are too complex or crash.
-

5. Corporate Customer (Trip Planning)

Name: Claire Laurent

Age: 41

Role: Executive assistant in a large company

Main Goals:

- Plan trips for **10 employees** every week.
- Track trip progress in real time.
- Have **direct contact with drivers** in case of issues.

Needs:

- A platform to:
 - Book recurring trips (e.g., every Monday at 8 AM for the CEO).
 - Receive **real-time notifications** (delays, cancellations).
 - Contact the driver directly if needed (e.g., address change).
- **Clear reporting** to justify expenses to accounting.

Frustrations:

- Drivers who don't arrive on time and make employees wait.
 - Lack of visibility on trip status (e.g., "Where is Mr. Dupont's taxi?").
 - Invoices arriving late or with errors.
-

D. Page Functionality

1. Login Page

- **Purpose:** Authenticate users and redirect them to their respective dashboards.
 - **Features:**
 - Login form with email and password fields.
 - "Forgot Password" link to reset password.
 - Role-based redirection (Admin, Driver, Customer).
 - **Actors:** All users.
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2. Admin Dashboard

- **Purpose:** Central hub for managing rides, drivers, customers, and invoices.
- **Features:**
 - **Ride Management:**
 - View, add, edit, and delete rides.
 - Assign rides to drivers manually or automatically.
 - Import rides via OCR (photos, text).
 - **Driver Management:**
 - View, add, edit, and deactivate driver accounts.
 - Validate driver documents (ID, license).
 - **Customer Management:**
 - View and manage customer accounts.
 - **Billing:**

- Generate, view, and send invoices.
 - Apply discounts and manage payment reminders.
 - **Reports & Statistics:**
 - Generate reports on rides, revenue, and driver performance.
 - Export data to CSV/PDF.
 - **Notifications:**
 - Send manual notifications to drivers or customers.
 - **Actors:** Admins.
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3. Driver Dashboard

- **Purpose:** Allow drivers to view and manage their assigned rides, availability, and earnings.
 - **Features:**
 - **Ride Management:**
 - View suggested rides based on location and availability.
 - Request assignment to a ride.
 - View ride history and details (departure, destination, price).
 - **Availability:**
 - Toggle availability status (available/unavailable).
 - **Notifications:**
 - Receive real-time alerts for new rides, assignments, and messages.
 - **Earnings & Ratings:**
 - View earnings, completed rides, and customer ratings.
 - Rate customers after completing a ride.
 - **Geolocation:**
 - View ride locations on an interactive map (Mapbox/Google Maps).
 - Enable live GPS tracking for customers (optional).
 - **Chat:**
 - Communicate with customers via in-app chat.
 - **Actors:** Drivers.
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4. Customer Dashboard

- **Purpose:** Allow customers to book rides, track trips, and manage payments.
- **Features:**
 - **Ride Booking:**
 - Book a ride manually (form with date, time, location, price).
 - View estimated fare before booking.
 - **Ride Tracking:**
 - Track assigned driver's location in real-time (GPS).
 - View ride history and receipts.
 - **Payments:**
 - Pay via app (Stripe/PayPal) or mark as paid in cash.
 - View and download invoices.
 - **Notifications:**
 - Receive real-time alerts for ride status (assigned, in progress, completed).

- **Driver Rating:**
 - Rate drivers after completing a ride (1-5 stars + comments).
 - **Support:**
 - Contact driver or admin via in-app chat.
 - **Actors:** Customers.
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5. Billing Page (Admin)

- **Purpose:** Manage invoicing, payments, and financial reporting.
 - **Features:**
 - **Invoice Generation:**
 - Create individual or grouped invoices from rides.
 - Automatic invoice numbering.
 - **Payment Tracking:**
 - Track payment status (unpaid, paid, cancelled).
 - Send automatic reminders for overdue payments.
 - **Discounts:**
 - Apply exceptional discounts to invoices (with justification).
 - **Export:**
 - Export invoices and financial reports to CSV/PDF.
 - **Actors:** Admins.
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6. Notifications Page (All Users)

- **Purpose:** Centralize and manage notifications for rides, invoices, and system alerts.
 - **Features:**
 - View all notifications (unread/read).
 - Filter notifications by type (ride assignments, invoices, urgent alerts).
 - Customize notification preferences (email, in-app, push).
 - **Actors:** All users.
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7. Profile Page (All Users)

- **Purpose:** Allow users to view and edit their personal information.
 - **Features:**
 - **Personal Information:**
 - Edit email, phone number, and password.
 - **Account Settings:**
 - Toggle notification preferences.
 - Deactivate account (admins only for drivers/customers).
 - **Activity Log** (Admins only):
 - View user activity (logins, actions).
 - **Actors:** All users.
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8. OCR Import Page (Admin/Drivers)

- **Purpose:** Import ride data from photos or text using OCR.
 - **Features:**
 - Upload photos of tickets or invoices.
 - Automatically extract ride details (date, time, location, price).
 - Manually validate or correct extracted data before saving.
 - **Actors:** Admins, Drivers.
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9. Admin Excel-like View

- **Purpose:** Provide a dynamic, spreadsheet-like interface for bulk ride and driver management.
 - **Features:**
 - **Dynamic Table:**
 - Sort, filter, and search rides/drivers by any field.
 - Bulk actions (assign, delete, export).
 - **Export:**
 - Export filtered data to CSV/PDF (limit: 10,000 rows).
 - **Actors:** Admins.
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10. Real-Time Chat (Drivers/Customers)

- **Purpose:** Facilitate communication between drivers and customers.
 - **Features:**
 - Send and receive text messages.
 - View conversation history (30-day retention).
 - Receive notifications for new messages.
 - **Actors:** Drivers, Customers.
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11. Geolocation & Maps (Drivers/Customers)

- **Purpose:** Provide real-time location tracking and route visualization.
 - **Features:**
 - **Drivers:**
 - View ride pickup/drop-off locations on a map.
 - Enable/disable live GPS sharing with customers.
 - **Customers:**
 - Track driver's real-time location during a ride.
 - View estimated arrival time.
 - **Actors:** Drivers, Customers.
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12. Calendar Integration (Drivers)

- **Purpose:** Sync ride schedules with personal calendars (Google/Outlook).
- **Features:**
 - Bidirectional sync: rides appear as calendar events.
 - Receive alerts for upcoming rides.

- **Actors:** Drivers.
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Notes:

- **Security:** All pages enforce role-based access control (e.g., drivers cannot access admin features).
 - **Responsiveness:** All pages are optimized for web and mobile use.
 - **Performance:** Critical actions (e.g., ride assignment, payment) load in < 2 seconds.
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Glossary

- **Admin Dashboard:** Central interface for administrators to manage rides, drivers, customers, and invoices. It provides tools for ride assignment, billing, reporting, and notifications.
- **Assignment Request:** A request made by a driver to be assigned to a specific ride. The status can be pending, approved, or rejected.
- **Better-Auth:** Authentication system used for secure user login and role management in the application.
- **Billing:** Process of generating, sending, and tracking invoices for rides. Includes features for discounts, reminders, and payment status updates.
- **Calendar Integration:** Feature that allows drivers to sync their ride schedules with personal calendars (e.g., Google Calendar, Outlook).
- **Customer Dashboard:** Interface for customers to book rides, track trips, manage payments, and communicate with drivers.
- **Driver Dashboard:** Interface for drivers to view and manage their assigned rides, availability, earnings, and ratings.
- **Dynamic Table:** Interactive table in the Admin Dashboard that allows filtering, sorting, and exporting data for rides and drivers.
- **Excel-like View:** Spreadsheet-like interface for bulk management of rides and drivers, supporting actions like sorting, filtering, and exporting.
- **Geolocation:** Integration with mapping services (e.g., Mapbox, Google Maps) to provide real-time location tracking and route visualization.
- **GDPR:** General Data Protection Regulation, a legal framework for personal data protection in Europe. Ensures data privacy and security.
- **Invoice:** Document generated for billing purposes, detailing ride costs, taxes, and payment status.
- **Live GPS Tracking:** Feature that allows customers to track the real-time location of their assigned driver during a ride.
- **Neon:** Serverless PostgreSQL service optimized for scalable and modern applications.
- **NextJS:** React framework used for building the web application, providing server-side rendering and static site generation.
- **Notifications:** Real-time alerts sent to users for events like new rides, assignments, invoices, and urgent updates.
- **OCR (Optical Character Recognition):** Technology used to extract text from images (e.g., tickets, invoices) for ride data import.
- **Partial Payment:** Feature allowing customers to pay a portion of the total invoice amount (minimum 30%).
- **Real-Time Chat:** In-app messaging system for communication between drivers and customers.

- **Ride:** A taxi trip with details like departure, destination, time, price, and status (pending, assigned, completed, cancelled).
- **Ride History:** Record of past rides for drivers and customers, including details like date, status, and price.
- **Ride Import:** Process of adding ride data via form, text, or photo (using OCR).
- **Ride Management:** Features for adding, editing, assigning, and tracking rides.
- **Ride Options:** Additional services (e.g., VIP, baby seat) that can be selected for a ride, with associated prices.
- **Ride Suggestions:** Algorithm-based recommendations for rides based on driver availability and location.
- **Role-Based Access Control:** System that restricts access to features based on user roles (Admin, Driver, Customer).
- **Shift Planning:** Management of driver shift schedules, including start/end times and status (planned, active, completed, cancelled).
- **Stripe/PayPal:** Secure online payment solutions integrated for processing ride payments.
- **Two-Factor Authentication (2FA):** Security method requiring two proofs of identity for user login.
- **User Management:** Features for registering, validating, modifying, and deactivating user accounts.
- **User Role:** Classification of users in the system (Admin, Driver, Customer) that determines access and permissions.
- **Vitest:** Testing framework used for unit and integration testing in the application.